

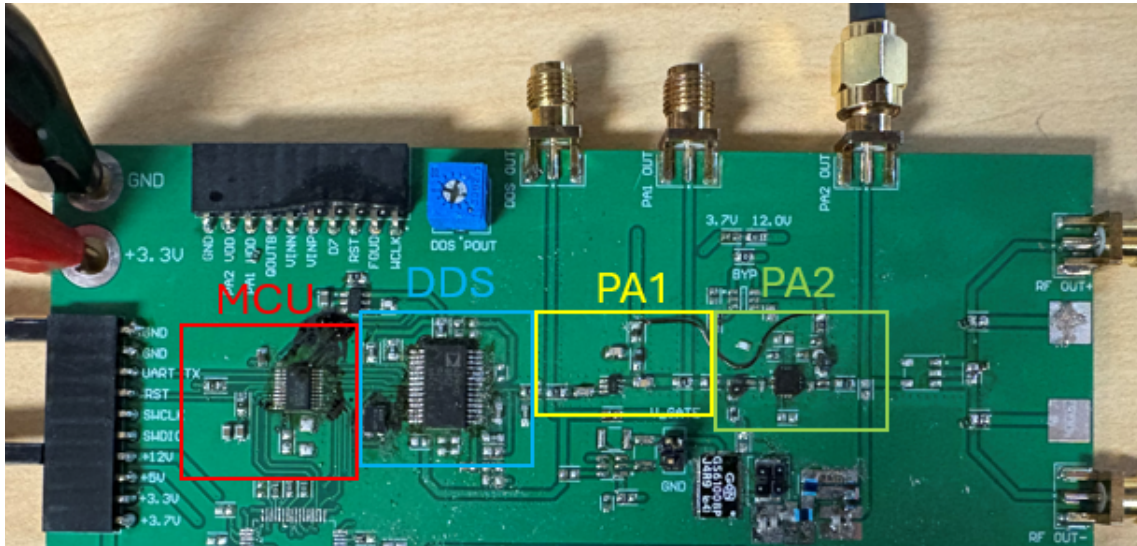
Beta Build

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Project repository: <https://github.com/jdados/dtcp>

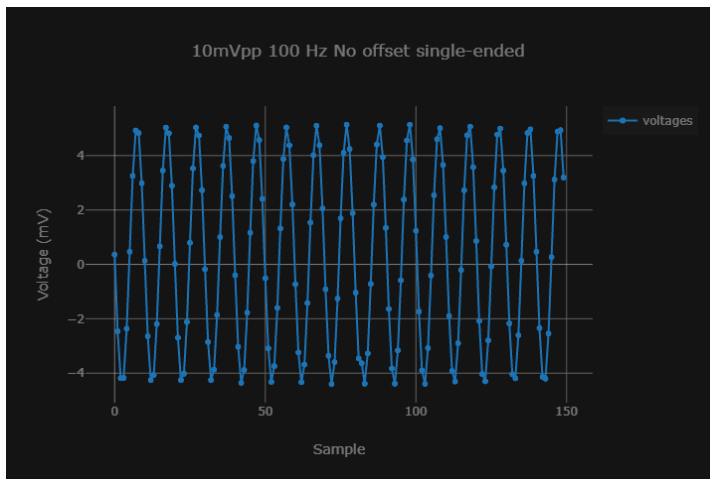
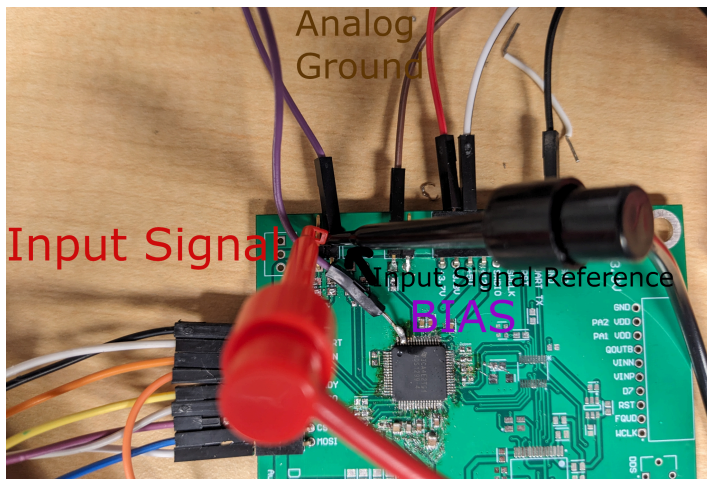
1. Current state of the project work

Since the alpha build, we've been working on integrating the RF transmitter part of the design into a single board. After many hours of soldering and re-soldering, we were able to produce and validate that circuit:



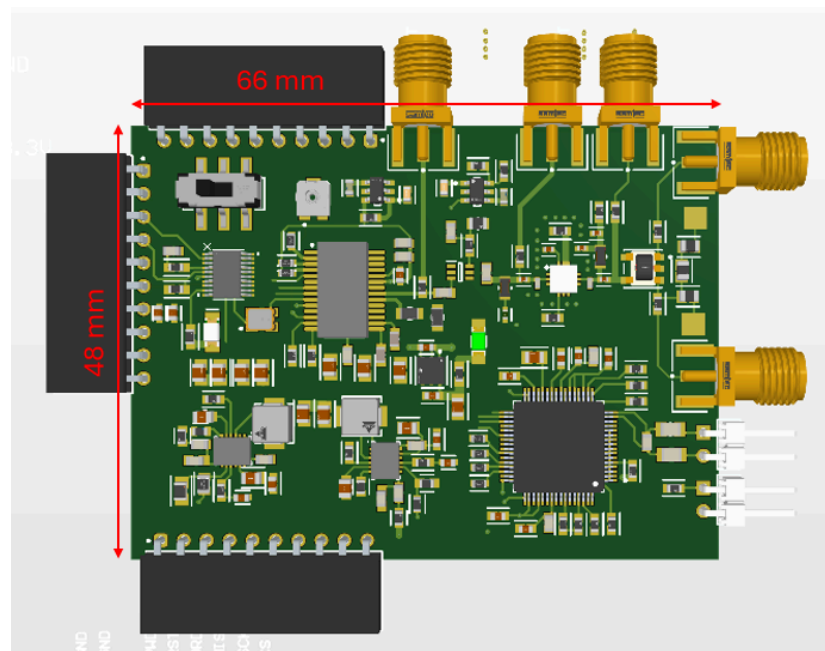
RF transmitter portion of the device (MCU + DDS + PA) integrated on one board

When it comes to the EMG recording circuit, most of our work has been focused on improving the firmware and circuit setup to obtain measurements in a realistic setting (no external bias, 10 mVpp AC signal):



Current setup for data measurement with the AFE and the readings from a signal generator

Finally, we started working on the final PCB design that captures all the learnings from the first PCB, delivering a complete system free of the bugs we identified so far:



3D view of the new PCB layout (work in progress)

2. Usability

a. Interface

UART-based interface is being developed to be included in the beta version.

b. Navigation and perception

Once the device is flashed with firmware, the only controls required are the power supply / battery connections and the output power setting potentiometer. Then, the board will start transmitting power to any 50 ohm load and performing the EMG measurements.

c. Responsiveness

Interaction of the user with the board will be expanded in the beta version by adding a power switch for the board to turn the whole system on and off.

3. Build quality

a. Robustness and Consistency

The beta version of our circuits eliminates all the bugs of the alpha version, providing a more robust and reliable system capable of continuous operation for at least 30 min.

b. Aesthetic rigor

Board layout aesthetics will be considered in the development of the beta PCB, because it will be used for the paper figures.

4. Vertical features

a. External interface

All the external interfaces are verified and carry out functionality correctly.

b. Persistent State

Currently, EMG recording data storage is only implemented on the microcontroller. As we develop the beta device, we'll add an option to save the recorded data on the PC.

c. Internal Systems

All the bugs and design errors have been identified and will be eliminated in the PCB design process for the beta device.